

Geocomputation & Machine Learning course

Quick recap

The meeting was the opening session of the Spatial Ecology Geocomputation Course 2024, led by Tushar Sethi and Giuseppe Amatulli, with presentations by Francesco and Rafael. Tushar welcomed the participants, outlined the course structure, and emphasized the use of Discord and video during classes. Giuseppe presented the course focus on geocomputation and machine learning, highlighting the importance of Linux, GDAL, and PKTools for data preparation and the need for reproducibility. Francesco discussed the concept of reproducible research and the use of free software and Debian GIS. Rafael introduced the machine learning component, explaining neural networks and supervised learning, and mentioned the use of PyTorch. Participants from various countries and disciplines introduced themselves, sharing their backgrounds and expectations for the course. The session also covered the installation and setup of the Ubuntu virtual machine, with Giuseppe guiding through the process and troubleshooting common issues.

Next steps

Giuseppe

- Add participants to the Discord channel once they join.
- Provide additional troubleshooting help via Discord for virtual machine installation issues.

Collaboration

- All participants: Send a photo to Giuseppe for the course roster.
- All participants: Set up Discord for course communications and use their real names.
- All participants: Install Ubuntu 26.04 LTS on a virtual machine (VirtualBox or UTM for Mac) and complete the software installations (GDAL, QGIS, PKTools, etc.) as per the instructions.

- All participants: Clone the course data repository (SEData) and set up a working folder in the shared folder.
- All participants: Attend the optional troubleshooting session tomorrow at 1 p.m. if experiencing issues with the virtual machine installation.
- All participants: Complete approximately 20 hours of homework per week after lectures.
- All participants: Select a project in collaboration with Giuseppe for the in-person workshop in Matera.
- All participants: Aim to complete the Geocomputation part and have a table ready for the machine learning part before the Matera workshop.

Summary

Spatial Ecology Geocomputation Course Overview

Tushar Sethi welcomed participants to the Spatial Ecology Geocomputation Course 2020 and introduced himself as the director alongside co-director Giuseppe. He outlined the course structure, mentioning that instructors Giuseppe, Rafaeli, and Francesco would be teaching, with Francesco joining later for an in-person workshop in Matera. Tushar provided course logistics, including the requirement to keep video on during sessions, complete 20 hours of homework weekly, and participate in a final hackathon project presentation in Matera to receive a completion certificate.

Geocomputation Course Introduction

Giuseppe Amatulli introduced the Geocomputation course, which he has been running for 20 years, focusing on geoenvironmental data processing using raster and vector data. The course will cover tools like GDAL, MKTools, and GRASS, with hands-on exercises in a Linux environment using virtual machines. The course is structured into two main parts: Geocomputation and Machine Learning, with a catch-up session in between, and participants are expected to complete geocomputation work before attending the in-person session in Matera.

Linux Geospatial Analysis Course Introduction

The meeting was an introduction session where Giuseppe welcomed participants to a Linux and geospatial analysis course. Participants introduced themselves, sharing their backgrounds, current research projects, and experience levels with Linux and related programming languages. Most participants, including Areeba, Chintia, Andrew, Abdul, Sebastian, Johannes, Lianda, Juicy, Ali, Eva, Katharina, Shiva, and Annabell, indicated they had limited to no experience with Linux but were interested in learning. A few participants, such as Cynthia and Eva, mentioned their experience with Python, R, and other programming languages relevant to their research. The session concluded with an introduction to Francesco, who could not attend fully.

Reproducible Research and Computing Environments

Francesco presented his background in remote sensing and geospatial computations, explaining his work on reproducible research and Debian GIS project. He outlined plans to discuss reproducible computer environments, data science components, and scaling applications for high-performance computing during an upcoming presentation in Matera. Giuseppe introduced the course's focus on Linux environments for reproducible research, highlighting their advantages for connecting different software tools and managing computational processes. The conversation ended with Giuseppe explaining the course's goal of developing self-learning skills in computational domains, with Rafael set to cover machine learning aspects in future sessions.

Machine Learning for Geodata Applications

Rafael presented an overview of machine learning applications for geodata, focusing on object detection from satellite images and other domain-specific tasks. He explained the evolution of neural networks from simple perceptrons to modern multi-layer perceptrons, and outlined the machine learning pipeline including data preprocessing, model training, and deployment. Rafael announced that the course will use PyTorch framework and requires basic knowledge of Python, algorithms, classical machine learning, linear algebra, and calculus, with additional support available for participants who need it.

Machine Learning Techniques Discussion

The meeting focused on discussing participants' experiences with machine learning techniques. Several participants, including Giuseppe, Abdul, Eva, Ali, Shiva, and Sun, shared their current use of various algorithms such as k-means clustering, random forest, XGBoost, LSTM, and self-

organizing maps in their research. Raffa assured participants that they would receive guidance on Python and machine learning code through dedicated sessions during the course. The session concluded with an introduction to additional learning resources, including a previous Geo Open Arc course on high-performance computing for geodata processing.

VirtualBox Ubuntu Installation Walkthrough

Giuseppe led a detailed walkthrough of virtual machine installation using VirtualBox and Ubuntu 26 LTS, explaining the process of setting up the virtual environment, configuring memory and processor settings, and managing shared folders between the host and guest operating systems. He emphasized the importance of proper configuration to avoid performance issues and data management best practices, including using shared folders to avoid increasing the VDI file size. Giuseppe encouraged students to complete the installation process and addressed common issues like screen size and permission problems, offering troubleshooting tips and suggesting a follow-up session for additional support if needed.

GIS Software Installation Technical Session

Giuseppe led a technical session on software installation and setup, guiding participants through installing Ubuntu 20.04 with GIS software including GDAL, PKTools, QGIS Python, and GRASS. He explained the use of JupyterLab as the primary editor for the course and demonstrated how to work with GitHub repositories by creating a local copy of the SE data in a separate folder to avoid conflicts with Git operations. The session addressed technical issues including screen sharing problems and installation procedures, with participants confirming that the shared folder and software installations were working properly.

Virtual Machine Troubleshooting Session

Giuseppe led a troubleshooting session for virtual machine installations, helping students resolve issues with passwords, shared folders, and software installations. He provided specific commands for fixing shared folder access problems and confirmed that Abdul's Ubuntu installation was working properly after downloading the required 7GB files. Giuseppe announced that the next class scheduled for 1 PM would focus on troubleshooting rather than new content, and outlined the upcoming curriculum which will cover Linux command line, file management, AWK for text manipulation, and finally GDAL operations for geospatial data over the next two weeks.